

MARGDARSHAK AI (■■■■■■■■■■ AI)

"From Classroom to Dream Career — Guided by AI"

Founder & Lead Developer: Mehak
Edition & Track: SIH 2024 Edition

Live Platform: margdarshak-ai-khaki.vercel.app
GitHub Source: github.com/Mehak237/margdarshak-ai

1. Executive Summary & Ground Realities

Over **1.5 million engineers and degree students graduate in India every year**. Yet, national employability studies (including the Wheebox India Skills Report) reveal that **less than 45% of students from Tier-2 and Tier-3 institutions** possess job-ready technical, algorithmic, and conversational proficiencies. This deficit stems from an acute **information, guidance, and economic divide**: students submit generic resumes lacking company ATS keywords, miss out on thousands of crores in available scholarships, and cannot afford commercial coaching.

The Core Innovation: Architected and engineered by Mehak for SIH 2024, **Margdarshak AI** is an all-in-one, zero-cost, browser-based career accelerator. It combines client-side NLP heuristics with Google Gemini 1.5 Flash to provide instant **Resume vs. Dream Job Gap Analysis across 60+ Indian employers**, **25+ verified scholarships** with an AI Statement of Purpose (SOP) writer, **real-time interactive voice mock interviews**, and **9 Indian languages**.

2. Full-Stack Hybrid Architecture & Data Flow

Margdarshak AI uses a **Client-First Hybrid Edge Architecture**. Browser APIs execute 100% of operations client-side—guaranteeing **0ms cold starts**, total candidate privacy (PDFs never leave local memory), and zero hosting expenses—while allowing seamless cloud sync with Google Gemini and multi-user databases.

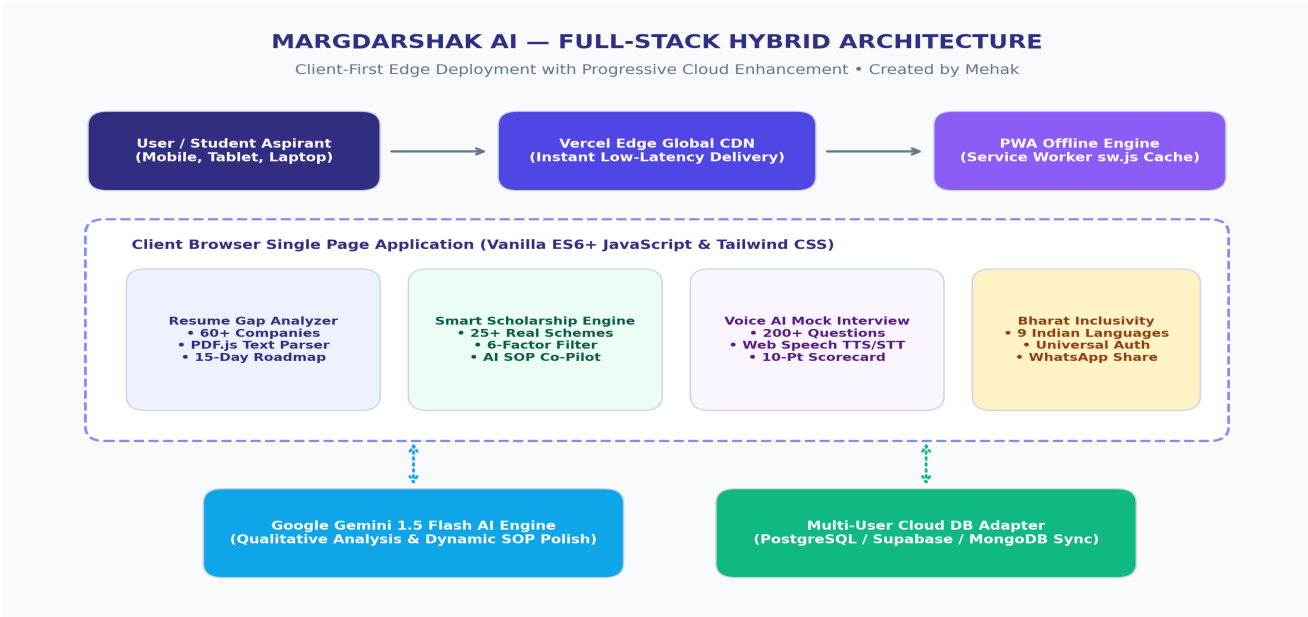
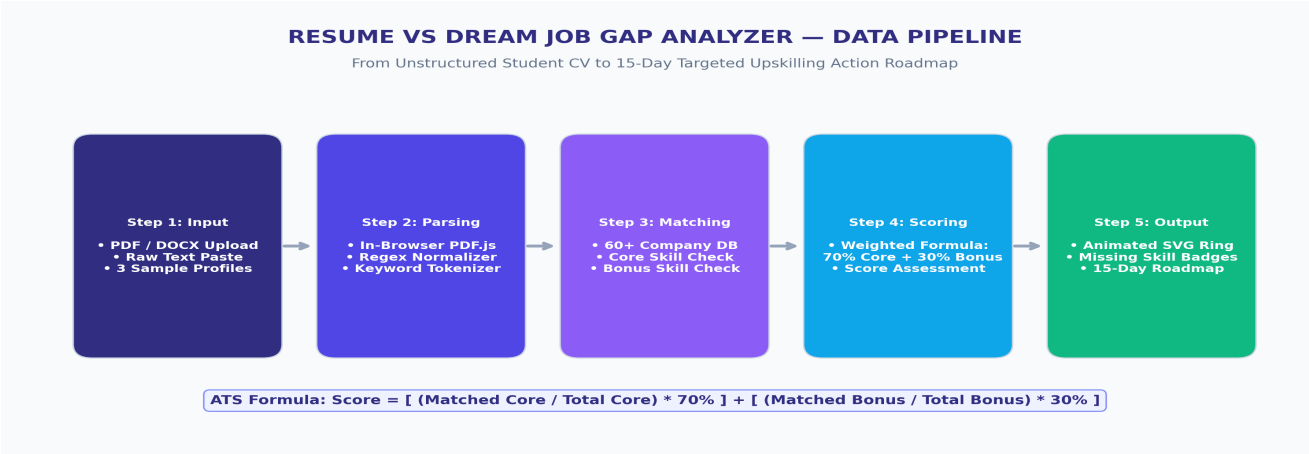


Figure 1: Full-Stack Hybrid Architecture — Client-First SPA + Progressive Cloud AI

3. Feature Deep Dive: Resume vs. Dream Job Gap Analyzer

The Gap Analyzer bridges the gap between candidate resumes and real-world hiring criteria. It features an integrated database of **60+ premier employers in India** (Google, Amazon, TCS Ninja/Digital, Infosys DSE, Wipro, Flipkart, Swiggy, Goldman Sachs) categorized into 5 competitive tiers.



5. Feature Deep Dive: Voice AI Mock Interviewer with Live Scorecard

Commercial coaching platforms charge Rs. 1,500 - Rs. 3,000 per mock session. Margdarshak AI delivers an interactive, voice-driven mock interview simulator 100% free right in the browser.

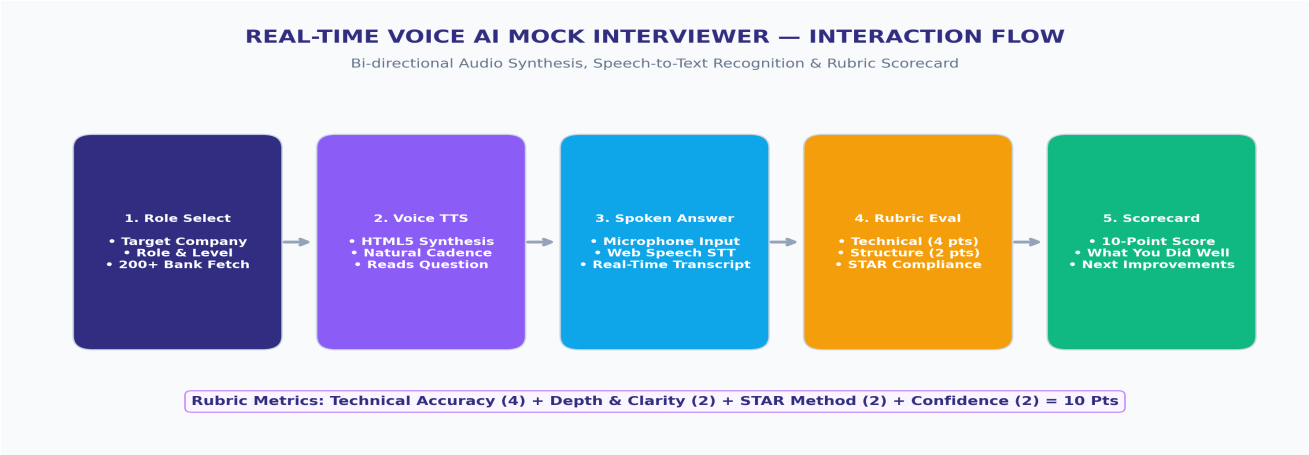


Figure 4: Bi-Directional Speech Synthesis (TTS) & Speech Recognition (STT) Interview Pipeline

- **Speech Synthesis (Voice Interviewer):** Vocalizes company-tailored interview questions using the HTML5 `SpeechSynthesisUtterance` API.
- **Speech-to-Text Recognition:** Listens to candidate answers via the browser microphone using `webkitSpeechRecognition`, transcribing speech into structured text.
- **10-Point Scorecard Rubric:** Evaluates answers across Technical Accuracy (4 pts), Clarity & Structure (2 pts), STAR Method Compliance (2 pts), and Delivery Confidence (2 pts), accompanied by actionable constructive improvement pointers.

6. Codebase Architecture & File Responsibilities

The project is structured with modular separation of concerns across single-page presentation, specialized JavaScript calculation engines, normalized datasets, and edge configurations.

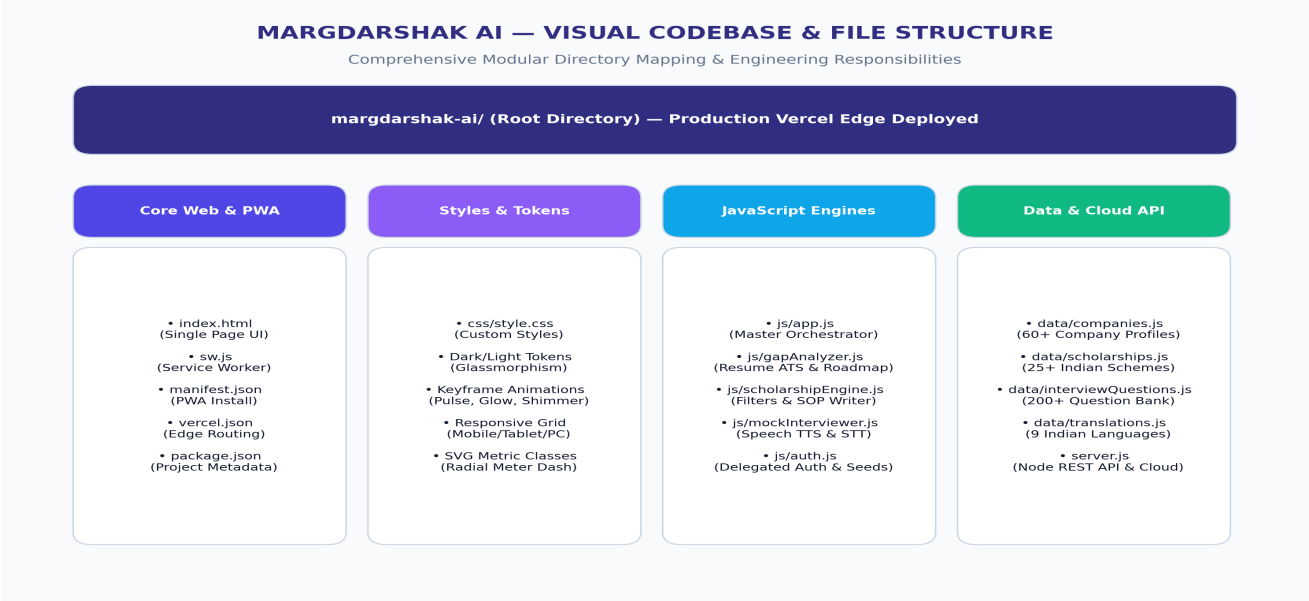


Figure 5: Visual Codebase Architecture & Modular File Responsibilities

7. Codebase Directory Inventory & Mathematical Model

File / Path	Module Role	Key Technical Functionality
index.html	SPA Markup	Full semantic single-page application UI, modular sections, navigation, and modals.
css/style.css	Styles & Theme	Custom dark/light glassmorphism design tokens, keyframe animations, and responsive utilities.
js/app.js	Master Controller	Client-side routing, tab coordination, theme switcher, and WhatsApp share handler.
js/gapAnalyzer.js	Gap Engine	Resume text extraction (PDF.js), 60+ company gap matcher, and 15-day roadmap builder.
js/scholarshipEngine.js	Scholarship Engine	Multi-factor scholarship query engine and committee-ready 4-paragraph AI SOP writer.
js/mockInterviewer.js	Voice Interviewer	Web Speech API voice synthesis, microphone STT, and 10-point scorecard rubric.
js/auth.js	Auth Controller	Universal document-delegated authentication, session management, and demo seeds.
data/companies.js	Dataset	60+ curated Indian tech companies with core & bonus skill definitions.
data/scholarships.js	Dataset	25+ verified Indian scholarships with live application links and eligibility rules.
data/translations.js	Dataset	Multilingual dictionary for 9 Indian languages (English, Hindi, Hinglish, etc.).
server.js	Backend API	Native Node.js REST API with cloud database support and Gemini proxy.
sw.js & manifest.json	PWA Engine	Service worker offline cache and web app installation manifest.
vercel.json	Cloud Config	Static edge routing and cloud production deployment configuration.

Mathematical ATS Scoring Formulation:

ATS Compatibility Score Formula:
Score_ATS = [(Matched_Core_Skills / Total_Core_Skills) x 70%] + [(Matched_Bonus_Skills / Total_Bonus_Skills) x 30%]

- **Score >= 80% (Green Tier):** Ready for Selection — Strongly aligned; ready to apply immediately.
- **Score 60% - 79% (Amber Tier):** Competitive with Gaps — Needs 5-10 days targeted gap closure using roadmap.
- **Score < 60% (Rose Tier):** Fundamental Foundations Required — Complete Phase 1 core curriculum.

8. Social Impact, NEP 2020 Alignment & Project Sign-Off

- **Viksit Bharat 2047 & NEP 2020:** Democratizes high-tier placement coaching for 10M+ Indian students in rural and non-metro colleges, breaking linguistic and economic barriers.
 - **Economic ROI:** Elevating a candidate from service baseline (Rs. 3.5 LPA) to a product/digital tier (Rs. 7.5 LPA) unlocks over **Rs. 25 Lakhs** in incremental lifetime earnings.
 - **Women in Tech Empowerment:** Actively flags high-value female schemes (AICTE Pragati) to boost gender equity in engineering.

PROJECT CERTIFICATION & MENTOR EVALUATION

This technical report certifies that **Margdarshak AI** has been conceptualized, architected, engineered, and deployed live to production by **Mehak** as an original technological innovation for the **Smart India Hackathon (SIH 2024 Edition)**.

Mehak
Founder & Lead Developer • Margdarshak AI
SIH 2024 Edition

Faculty / Mentor Signature
Academic Project Evaluator / SIH Committee
Department of Computer Science & Engineering